General Dessert #2: Power, Bandwidth, and Digital Protocols

Transmitter power limits across HF bands, the 60-meter band's special rules, and what's required before using a new digital protocol.

Group G1C · 8 pool questions · Self-study reference · Practice exams at eham.net/exams

Key Facts Reference

Transmitter Power Limits

Question	Answer	Q
Max power on 10.140 MHz (30 m band)	**200 watts PEP output.** 30 m has a special low limit.	G1C01
Max power on the 12-meter band	**1500 watts PEP output.**	G1C02
Max power on the 28 MHz (10 m) band for General	**1500 watts PEP output.**	G1C05
Max power on the 1.8 MHz (160 m) band	**1500 watts PEP output.**	G1C06
What measurement FCC rules regulate	**PEP output from the transmitter.**	G1C11

60-Meter Band Rules

Question	Answer	Q
Max bandwidth for USB on 60 m	**2.8 kHz.**	G1C03
FCC requirement when operating on 60 m	**If using an antenna other than a dipole, keep a record of antenna gain.**	G1C04

New Digital Protocols

Question	Answer	Q
Required before using a new digital protocol on the air	**Publicly document the technical characteristics of the protocol.**	G1C07

Quick mental map: The General-class HF/MF power limit is **1500 W PEP output** on most bands, with two exceptions: **30 m is capped at 200 W PEP output**, and **60 m is limited to 100 W ERP relative to a half-wave dipole**. The 60-m band has unique rules: 2.8 kHz max bandwidth on USB, and you must log antenna gain if you use anything other than a dipole. New digital modes are legal as long as their technical details are public.

Dessert 2 Question Bank (G1C)

G1C01. What is the maximum transmitter power an amateur station may use on 10.140 MHz?
A. 200 watts PEP output
B. 1000 watts PEP output
C. 1500 watts PEP output
D. 2000 watts PEP output

G1C02. What is the maximum transmitter power an amateur station may use on the 12-meter band?
A. 50 watts PEP output
B. 200 watts PEP output
C. 1500 watts PEP output
D. An effective radiated power equivalent to 100 watts from a half-wave dipole

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G1C03. What is the maximum bandwidth permitted by FCC rules for amateur radio stations transmitting on USB frequencies in the 60-meter band?
A. 2.8 kHz
B. 5.6 kHz
C. 1.8 kHz
D. 3 kHz

G1C04. Which of the following is required by the FCC rules when operating in the 60-meter band?
A. If you are using an antenna other than a dipole, you must keep a record of the gain of your antenna
B. You must keep a record of the date, time, frequency, power level, and stations worked
C. You must keep a record of all third-party traffic
D. You must keep a record of the manufacturer of your equipment and the antenna used

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G1C05. What is the limit for transmitter power on the 28 MHz band for a General Class control operator?
A. 100 watts PEP output
B. 1000 watts PEP output
C. 1500 watts PEP output
D. 2000 watts PEP output

G1C06. What is the limit for transmitter power on the 1.8 MHz band?
A. 200 watts PEP output
B. 1000 watts PEP output
C. 1200 watts PEP output
D. 1500 watts PEP output

G1C07. What must be done before using a new digital protocol on the air?
A. Type-certify equipment to FCC standards
B. Obtain an experimental license from the FCC
C. Publicly document the technical characteristics of the protocol
D. Submit a rule-making proposal to the FCC describing the codes and methods of the technique

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G1C11. What measurement is specified by FCC rules that regulate maximum power?
A. RMS output from the transmitter
B. RMS input to the antenna
C. PEP input to the antenna
D. PEP output from the transmitter

Answer Key

G1C01: A G1C02: C G1C03: A G1C04: A

G1C05: C G1C06: D G1C07: C G1C11: D